

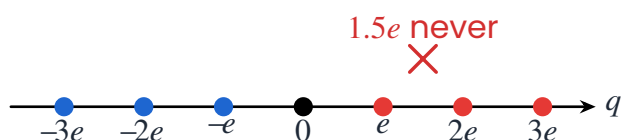
ELECTROSTATICS

Charge • Coulomb's Law

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1 Properties of Charge

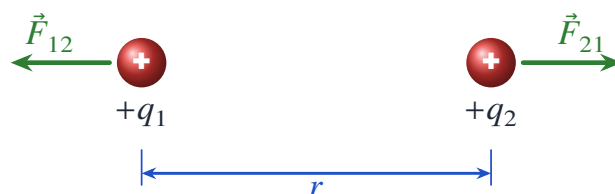
$$q = \pm ne, \quad e = 1.6 \times 10^{-19} \text{ C}$$



Quantised, conserved and additive:
only whole multiples of e exist.

2 Coulomb's Law

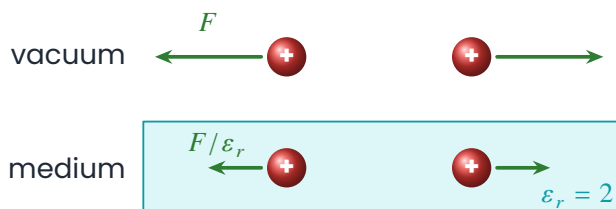
$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1q_2|}{r^2}$$



$1/4\pi\epsilon_0 = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$; equal and
opposite, along the line joining them.

3 In a Medium

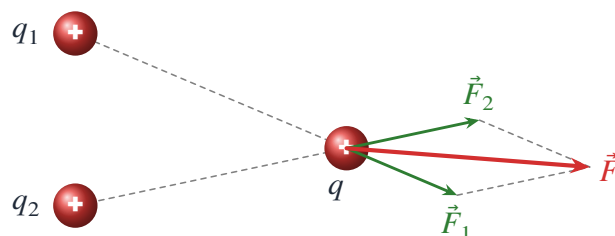
$$F_{\text{medium}} = \frac{F_{\text{vacuum}}}{\epsilon_r}$$



$\epsilon_r = \epsilon/\epsilon_0 \geq 1$ (dielectric constant);
water ≈ 80 .

4 Superposition

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \dots$$



Each pair acts as if the others were
absent; add the forces as vectors.

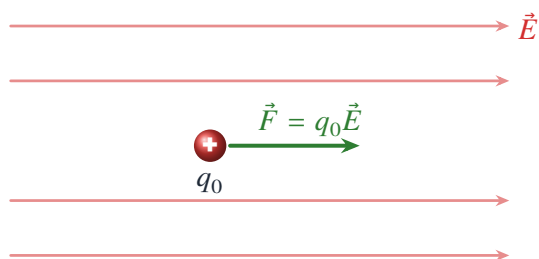
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Electric Field

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5 Electric Field

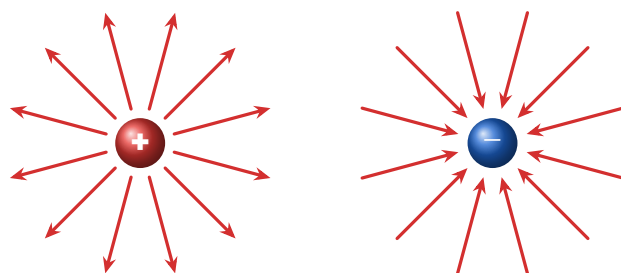
$$\vec{E} = \frac{\vec{F}}{q_0} \quad (q_0 \rightarrow 0)$$



Unit $\text{N C}^{-1} = \text{V m}^{-1}$. On any charge q :
 $\vec{F} = q\vec{E}$.

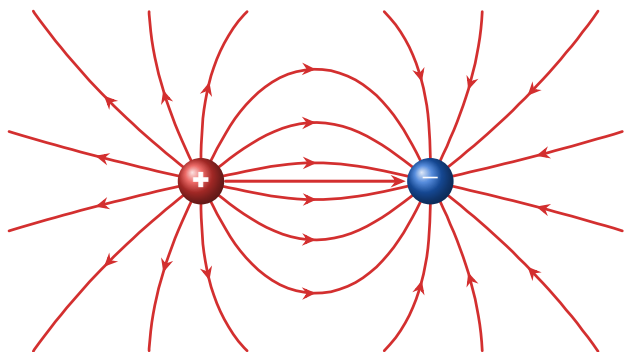
6 Field of a Point Charge

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$



Radially out of $+q$, into $-q$; falls as $1/r^2$.

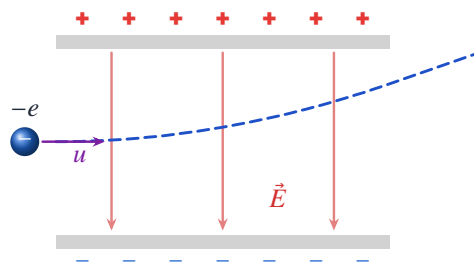
7 Field Lines



Start on $+$, end on $-$; never cross;
crowded where E is strong; meet a
conductor at 90° .

8 Charge in a Uniform Field

$$a = \frac{qE}{m}, \quad y = \frac{qE}{2mu^2} x^2$$



A parabola, like a projectile; an
electron bends towards the $+$ plate.

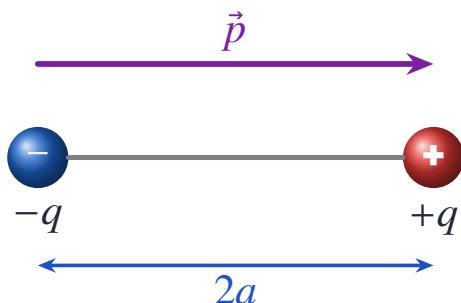
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Electric Dipole

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9 Dipole Moment

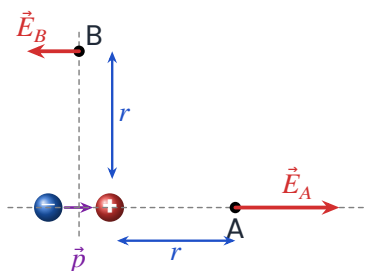
$$\vec{p} = q(2\vec{a})$$



Points from $-q$ to $+q$; unit C m. Net charge zero.

10 Axis vs Equator

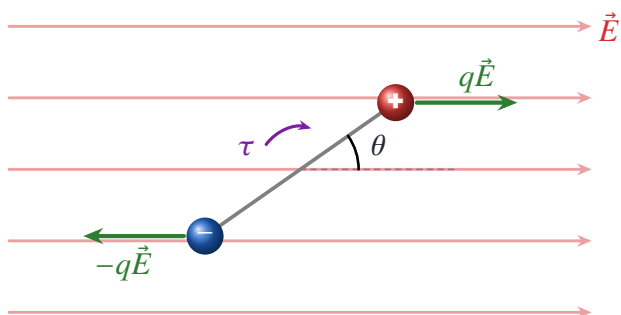
$$E_{\text{axis}} = \frac{2kp}{r^3}, \quad E_{\text{eq}} = \frac{kp}{r^3}$$



Far away, same r : axial field is double, along $\vec{p}$; equatorial is against $\vec{p}$.

11 Torque in a Uniform Field

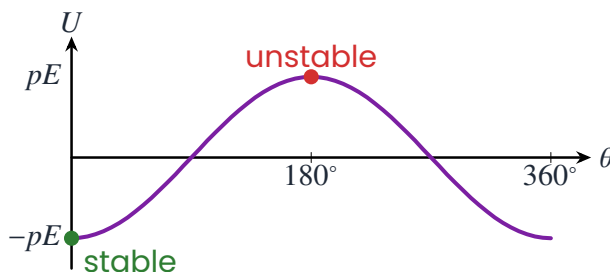
$$\vec{\tau} = \vec{p} \times \vec{E}, \quad \tau = pE \sin \theta$$



Net force zero; the couple turns $\vec{p}$ towards $\vec{E}$.

12 Potential Energy

$$U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$$



Work to turn it from θ_1 to θ_2 :
 $W = pE(\cos \theta_1 - \cos \theta_2)$.

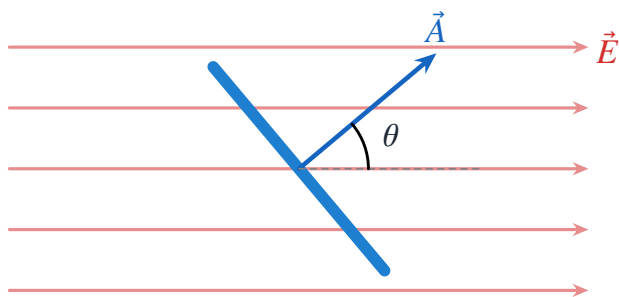
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Flux • Gauss's Law

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13 Electric Flux

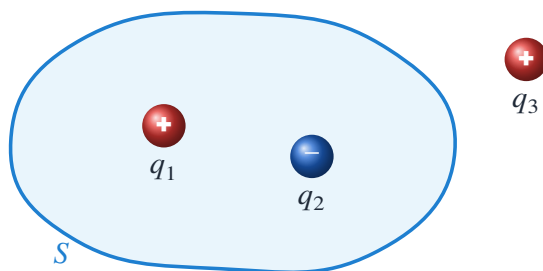
$$\phi = \vec{E} \cdot \vec{A} = EA \cos \theta$$



A scalar; unit $\text{N m}^2 \text{C}^{-1}$ ($= \text{V m}$).

14 Gauss's Law

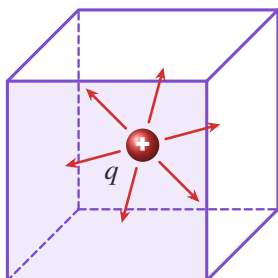
$$\oint_S \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$



Only $q_1 + q_2$ count. q_3 adds no net flux, but it does change $\vec{E}$ on S .

15 Charge at a Cube's Centre

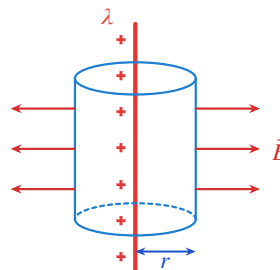
$$\phi_{\text{face}} = \frac{q}{6\epsilon_0}$$



At a corner: $q/8\epsilon_0$ through the cube,
 $q/24\epsilon_0$ through each face.

16 Infinite Line Charge

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$



Radially outward; no flux through the flat ends.

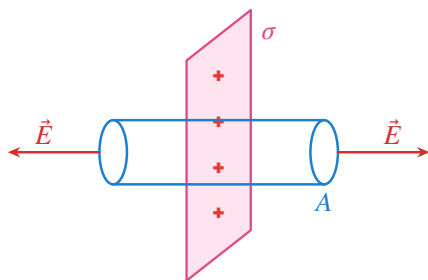
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Sheets • Conductors • Shells

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17 Infinite Plane Sheet

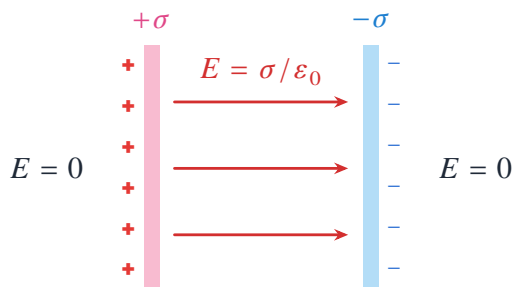
$$E = \frac{\sigma}{2\epsilon_0}$$



Perpendicular to the sheet; the same at every distance.

18 Two Opposite Sheets

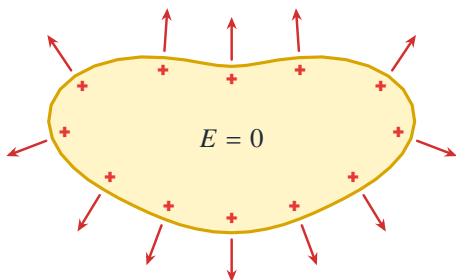
$$E_{\text{between}} = \frac{\sigma}{\epsilon_0}, \quad E_{\text{outside}} = 0$$



The two fields add between the sheets and cancel outside.

19 Charged Conductor

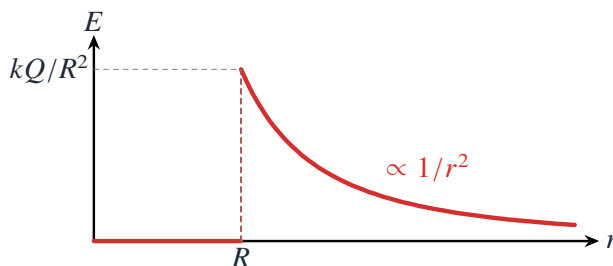
$$E_{\text{inside}} = 0, \quad E_{\text{surface}} = \frac{\sigma}{\epsilon_0}$$



All the charge sits on the outer surface; E leaves it at 90° .

20 Spherical Shell

$$E = \frac{kQ}{r^2} \quad (r \geq R), \quad E = 0 \quad (r < R)$$



Outside, the shell acts like a point charge at its centre.

ELECTROSTATICS

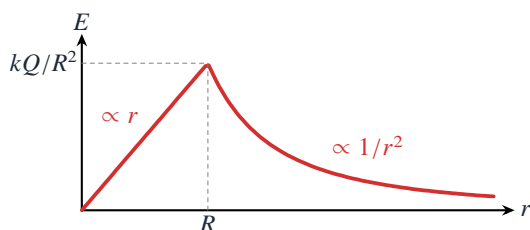
Sphere • Ring • Quick Numericals

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21 Uniform Solid Sphere

$$E = \frac{kQr}{R^3} \quad (r < R)$$

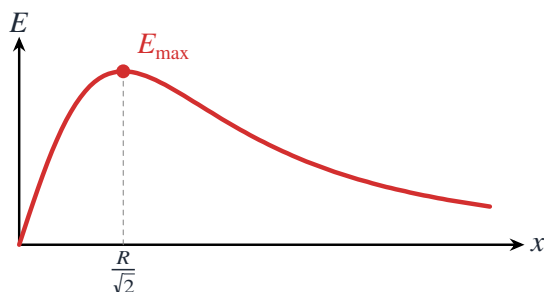
$$E = \frac{kQ}{r^2} \quad (r \geq R)$$



Largest at the surface.

22 Ring: Field on the Axis

$$E = \frac{kQx}{(R^2 + x^2)^{3/2}}$$

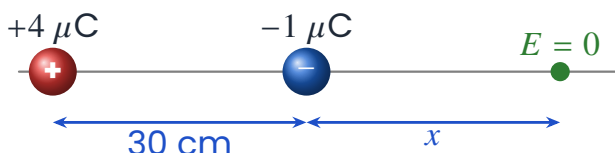


Zero at the centre, largest at $x = R/\sqrt{2}$.

23 Quick Numerical: $E = 0$

$+4 \mu\text{C}$ and $-1 \mu\text{C}$ are 30 cm apart.
Where is $E = 0$?

$$\frac{4}{(30 + x)^2} = \frac{1}{x^2} \Rightarrow 30 + x = 2x$$



$x = 30 \text{ cm}$ beyond the $-1 \mu\text{C}$

24 Quick Numerical: Torque

A dipole $p = 4 \times 10^{-9} \text{ C m}$ makes 30°
with $E = 5 \times 10^4 \text{ N C}^{-1}$. Find τ and U .

$$\tau = pE \sin 30^\circ = 1.0 \times 10^{-4} \text{ N m}$$

$$U = -pE \cos 30^\circ = -1.73 \times 10^{-4} \text{ J}$$

ELECTROSTATICS

Quick Revision • Traps

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25 Field of Every Distribution

Charge	Field outside / at r	Inside
Point charge q	kq/r^2	--
Line λ	$\lambda/2\pi\epsilon_0 r$	--
Sheet σ	$\sigma/2\epsilon_0$	--
Conductor (surface σ)	σ/ϵ_0	0
Shell Q, R	kQ/r^2	0
Solid sphere Q, R	kQ/r^2	kQr/R^3
Ring, on its axis	$kQx/(R^2 + x^2)^{3/2}$	0 at centre

26 Common Traps

- Field lines never cross: $\vec{E}$ has one direction at a point.
- Flux depends on q_{enc} only; $\vec{E}$ on S depends on every charge.
- An electron accelerates against $\vec{E}$.
- In a non-uniform field a dipole feels a net force too.

27 Constants & Units

Quantity	Value
e	$1.6 \times 10^{-19} \text{ C}$
ϵ_0	$8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$
$1/4\pi\epsilon_0$	$9 \times 10^9 \text{ N m}^2\text{C}^{-2}$
E	$\text{N C}^{-1} = \text{V m}^{-1}$
$\phi ; p$	$\text{N m}^2\text{C}^{-1} ; \text{C m}$